

Chapter 1

Math review

1.1 Mathematical Logic and proof writing.

In this section we study basic mathematical logic and some elements of proof writing.

1.1.1 Logical operators:

A “proposition” is a statement that is either true or false, but never both. In an abstract sense economics and mathematics look for “propositions” that are true and say something about the objects studied by each discipline. Then, it is worth studying the construction of propositions from more basic ones. Consider a pair of propositions p, q . Examples are: “John is tall,” “ $0 > 1$,” “market is efficient,” etc. From propositions like p and q we can construct new propositions using logical operators. Table 1.1 shows the basic logical operators.

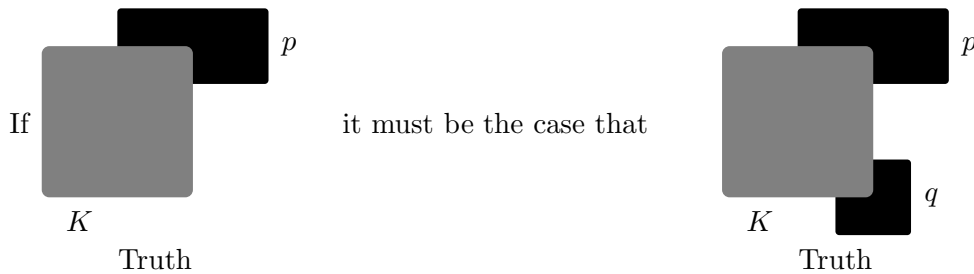
Even though it seems a simple task, constructing the negation of a composed statement is not trivial.

Exercise 1. *Construct the negation, in a non-trivial way, of the following propositions.*

1. $\exists x \in \mathbb{R} \text{ s.t. } x^2 = 2.$
2. $\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall n \geq N, |x_n - x| < \varepsilon.$
3. $\exists \varepsilon > 0 \text{ s.t. } \forall N \in \mathbb{N}, \exists n \in \mathbb{N}, n \geq N \text{ s.t. } |x_n - x| \geq \varepsilon.$

Operator	Notation	Proposition is true if:
Negation	$\neg p$	p is not true
and	$p \wedge q$	p and q are both true
or	$p \vee q$	either p or q is true
there exists	$\exists x \text{ s.t } p(x)$	there exists x that satisfies $p(x)$
there exists a unique	$\exists! x \text{ s.t } p(x)$	there exists a unique x that satisfies $p(x)$
for all	$\forall x, p(x)$	all x satisfy $p(x)$
implies	$p \Rightarrow q$	$(\neg p) \vee q$ is true
if and only if	$p \Leftrightarrow q$	$(p \Rightarrow q) \wedge (q \Rightarrow p)$ is true.

Table 1.1: Logical operators.

Figure 1.1: $p \Rightarrow q$.

1.1.2 Proof writing

Statements that are accepted as true are called **axioms**. All propositions that can be deduced from axioms are called **theorems**. Let us call your knowledge, K , the axioms plus all the theorems that you know are true.¹ Big part of your graduate work consists of providing the proof that a given proposition is true using your knowledge. That is, proving that such a proposition is a theorem. For the sake of concreteness consider a theorem of the form $p \Rightarrow q$. If the theorem $p \Rightarrow q$ is not trivial, then it is not in your knowledge. To understand what a proof of $p \Rightarrow q$ is, we must understand what $p \Rightarrow q$ means: if we add p to all that we know is true, i.e., K , then q should be true too. We illustrate this in Figure 1.1.²

¹We warn the reader that our definition of knowledge does not coincide with the standard definition of knowledge in economics.

²Thanks to Jorge Cossio, who showed me how to understand the structure of a proof using diagrams as in Figures 1 to 4.

There are some ways to provide a proof that $p \Rightarrow q$ is true (of course only if $p \Rightarrow q$ is really true). Let us describe some of them.

Direct proof

A direct proof exhibits a chain of propositions $s_1, s_2, \dots, s_n$ with the following properties: (1) for each $j = 1, \dots, n$, $\{K, P, s_1, \dots, s_{j-1}\}$ imply s_j , and (2) $s_n = q$.

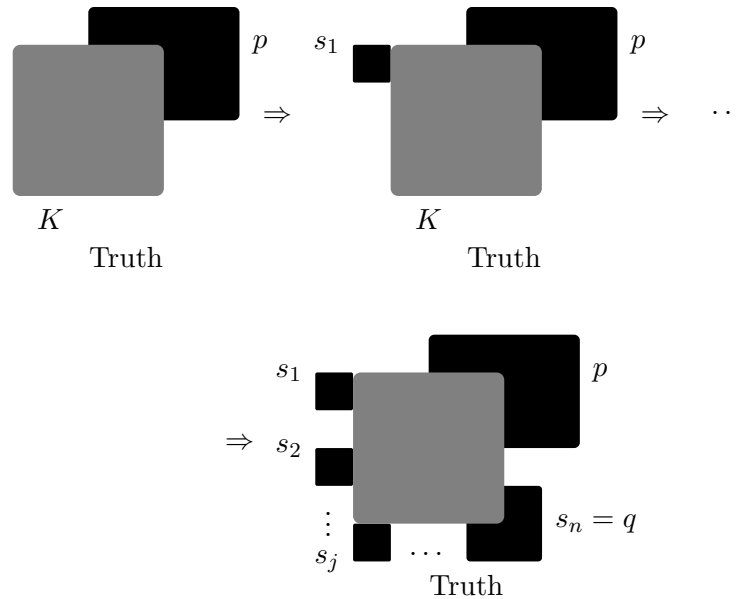


Figure 1.2: Direct proof.

A direct proof presents a chain of consequences of K plus the hypothesis p in such a way that at each step a proposition that is true is added, reaching finally the desired thesis q .

Example 1. Let m, r be integers. Prove “ $(m \text{ is even}) \Rightarrow (mr \text{ is even})$.”

Indirect proof

Recall that we are trying to prove $p \Rightarrow q$, that is, if we add p to the propositions that are true, then q must be true. There are just two possibilities: (1) q is true, and (2) q is not true, i.e., $\neg q$. Therefore one way to prove $p \Rightarrow q$ is to prove $\neg(p \wedge \neg q)$, that is, it is not true that p and $\neg q$ are true.

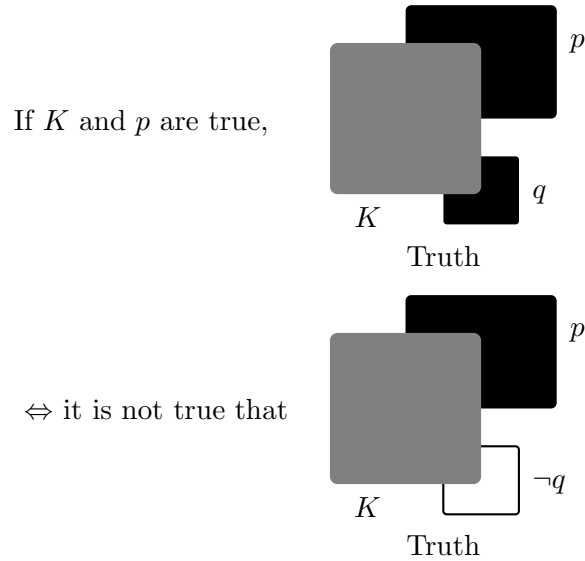


Figure 1.3: $(p \Rightarrow q) \Leftrightarrow \neg(p \wedge \neg q)$.

To prove $\neg(p \wedge \neg q)$ we proceed in the following way. First, suppose $(p \wedge \neg q)$ is true, that is, add p and $\neg q$ to what you know is true, K . Call K plus p and $\neg q$, the new truth NT . Second, provide a direct proof that $NT \Rightarrow \neg t$, where t is a proposition in NT . It is impossible that t and $\neg t$ are both true. Therefore, NT can not be true, and we proved that p and $\neg q$ can not be true together, that is, $\neg(p \wedge \neg q)$ (See Figure 1.4 for an illustration).

Indirect proofs are also known as “proofs by contradiction” or “reduction to the absurd.” Note that we did not impose any restriction on the final proposition that obtains the contradiction. That is, t in NT may be a proposition in K , or p , or even in $\neg q$. It is standard in the literature in mathematics and economics to reserve the term “proof by contradiction” for those proofs that arrive to a contradiction in K . When the final contradiction is obtained in p the proof is referred to as a “proof by contrapositive.” There are also proofs that obtain a contradiction in $\neg q$. They are logically correct. Nevertheless, some times, the contradiction of $\neg q$ indicates that you are able to write a direct proof instead. It is a matter of style, but you can take as a general rule that if you can write a direct proof, you should not write an indirect one.

We will use the term “contradiction” to refer to a proof that arrives to a contradiction in $K \cup \{q\}$, and “contrapositive” to refer to one that arrives

to a contradiction in p .

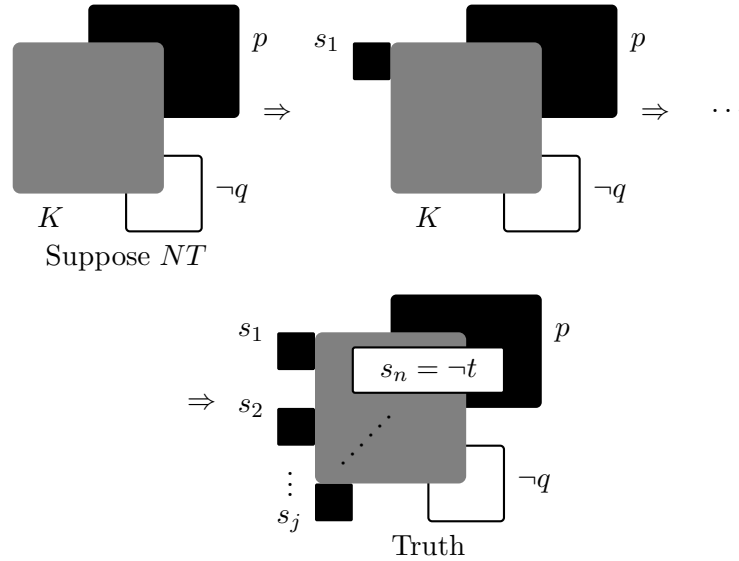


Figure 1.4: Indirect proof.

A proof by contradiction of $p \Rightarrow q$ is logically equivalent to a contrapositive. There is no sense in which a proposition is *more true* if it is proved in a different way. Nevertheless, there is a sense in which a proof by contrapositive is more desirable than a contradiction. A contrapositive may illuminate better the connection between the hypothesis p and the thesis q (see the Introduction in Royden [Royden \(1988\)](#) for a short digression on this point).

Example 2. Let r be an integer. Prove “ $(r^2 \text{ is odd}) \Rightarrow (r \text{ is odd})$.”

Example 3. Let $a, b \in \mathbb{R}$. Prove that if $\forall \varepsilon > 0, \varepsilon \in \mathbb{R}, a < b + \varepsilon$, then $a \leq b$.

Proof by induction

Consider a family of propositions $\{p(n)\}_{n \in \mathbb{N}}$, that is, a family of propositions indexed with the natural numbers. For instance, $\{2^n > n\}_{n \in \mathbb{N}}$. A proof by induction is a procedure only available when, as in this case, you can index the family of propositions with natural numbers. It is used to prove

that all propositions in the family are true, and proceeds as follows: (1) provide a proof that $p(1)$ is true, and (2) provide a proof of the proposition “ $\forall n \in \mathbb{N}, p(n) \Rightarrow p(n+1)$.”

Let us practice our proof writing skills and prove that (1) and (2) above provide a proof that $\forall n \in \mathbb{N}, p(n)$. We write this formally as a theorem.

Theorem 2. *Let $\{p(n)\}_{n \in \mathbb{N}}$ be a family of propositions. Then, $\{p(1) \text{ and } (\forall n \in \mathbb{N}, p(n) \Rightarrow p(n+1))\} \Rightarrow \{\forall n \in \mathbb{N}, p(n)\}$.*

Proof. We provide a direct proof of the proposition. Suppose $\{p(1) \wedge (\forall n \in \mathbb{N}, p(n) \Rightarrow p(n+1))\}$ is true. We want to prove $\{\forall n \in \mathbb{N}, p(n)\}$ is true too. Let $m \in \mathbb{N}$. We want to prove $p(m)$ is true. From the hypothesis we know it is true that $\forall n \in \mathbb{N}, p(n) \Rightarrow p(n+1)$. In particular for $n = 1$, we have $p(1) \Rightarrow p(2)$. As $p(1)$ is true then $p(2)$ is also true. Similarly as $p(2)$ is true then $p(3)$ is true too. Repeating this argument $m - 1$ times, we have $p(m)$ is true. As $m \in \mathbb{N}$ was arbitrary, then $\{\forall n \in \mathbb{N}, p(n)\}$. $\square$

Example 4. *Prove $\forall n \in \mathbb{N}, 2^n > n$.*

Proof. (1) $2 > 1$, (2) if $2^n > n$ then $2^{n+1} > 2n = n + n \geq n + 1$. $\square$

Example 5. *Prove $\forall n \in \mathbb{N}, 1 + 2 + \cdots + n = \frac{1}{2}n(n+1)$.*

General recommendations in proof writing

If you have to prove a proposition of the form $p \Leftrightarrow q$, recall that it is equivalent to prove the two propositions $p \Rightarrow q$ and $q \Rightarrow p$. You can provide the proof of both propositions independently. Make sure you are really proving the right propositions. A common mistake is to provide a direct proof of one of the propositions and then provide a proof by contradiction of the same proposition. As the proofs may have different nature, you may trick yourself and think you are providing a proof of the if and only if proposition.

After you learn how to write proofs, it is necessary to learn how to write in such a way that other people understand your proofs. This is not a trivial part of the process. Perhaps as difficult and for sure as important as the first one. Thomson (2011) provides a neat treatment of the topic especially directed to economists.

1.1.3 Problems

1. Provide a proof of the proposition in Example 3.

2. Provide a proof of the proposition in Example 5.
3. Show that $\sqrt{2}$ is not a rational number. (You may use the fact that any rational number can be expressed as p/q , where both p and $q \neq 0$ are integers, and at least one of them is an odd integer).
4. Using mathematical induction, prove the following:

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

5. Let $r \neq 1$. Using mathematical induction, prove that for each $n \in \mathbb{N}$,

$$1 + r + r^2 + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r}.$$

1.2 Sets, Functions, and binary relations

1.2.1 Sets

Basic Notation

- A : set

- $a \in A$: a is an element of A .
- $a \notin A$: a is not an element of A .

- Examples:

$$\begin{aligned} \mathbb{N} &= \{1, 2, 3, \dots\} \\ \mathbb{Z} &= \{\dots, -2, -1, 0, 1, 2, \dots\} \\ \mathbb{Q} &= \left\{x = \frac{p}{q} : p, q \in \mathbb{Z} \text{ and } q \neq 0\right\} \\ \mathbb{R} &: \text{the set of all real numbers} \\ \mathbb{R}_+ &= \{x \in \mathbb{R} : x \geq 0\} \\ \mathbb{R}_{++} &= \{x \in \mathbb{R} : x > 0\} \\ \emptyset &: \text{empty set (the set having no elements)} \end{aligned}$$

- $A \subset B \stackrel{\text{def}}{\Leftrightarrow} \forall a \in A, a \in B$ (Every element of A is also an element of B .)
- $A = B \stackrel{\text{def}}{\Leftrightarrow} A \subset B \text{ and } B \subset A$
- $A \subsetneq B \stackrel{\text{def}}{\Leftrightarrow} A \subset B \text{ and } A \neq B$.

Binary Operation of Sets

- Union:

$$A \cup B \equiv \{x : x \in A \text{ or } x \in B\}$$

- Intersection:

$$A \cap B \equiv \{x : x \in A \text{ and } x \in B\}$$

Definition 1. If $A \cap B = \emptyset$, A and B are disjoint.

- Difference:

$$A \setminus B \equiv \{x : x \in A \text{ and } x \notin B\}$$

- Complement:

$$\begin{aligned} X &: \text{ universal set} \\ A^c &\equiv X \setminus A \end{aligned}$$

- Symmetric difference:

$$A \Delta B \equiv (A \setminus B) \cup (B \setminus A)$$

It is convenient to use figures to represent sets. The most common diagrams used to represent sets are Venn diagrams. In these diagrams sets are represented as bubbles.

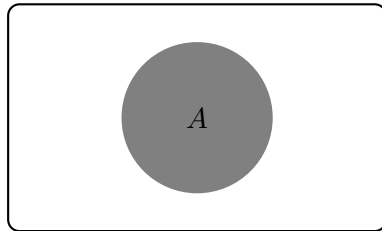


Figure 1.5: Venn diagram of set A .

Properties

1.

$$A \cap A = A, A \cup A = A, A \cup \emptyset = A, A \cap \emptyset = \emptyset.$$

2. Commutative laws:

$$\begin{aligned} A \cap B &= B \cap A; \\ A \cup B &= B \cup A. \end{aligned}$$

3. Associative laws:

$$\begin{aligned} (A \cap B) \cap C &= A \cap (B \cap C); \\ (A \cup B) \cup C &= A \cup (B \cup C). \end{aligned}$$

4. Distributive laws:

$$\begin{aligned} A \cap (B \cup C) &= (A \cap B) \cup (A \cap C); \\ A \cup (B \cap C) &= (A \cup B) \cap (A \cup C). \end{aligned}$$

5.

$$A \setminus B = A \cap B^c.$$

6.

$$(A \cap B) \cap (A \setminus B) = \emptyset.$$

7. DeMorgan's laws:

$$\begin{aligned} A \setminus (B \cup C) &= (A \setminus B) \cap (A \setminus C); \\ A \setminus (B \cap C) &= (A \setminus B) \cup (A \setminus C). \end{aligned}$$

Note: If $A = X$ (universal set), then DeMorgan's laws imply

$$\begin{aligned} (B \cup C)^c &= B^c \cap C^c; \\ (B \cap C)^c &= B^c \cup C^c. \end{aligned}$$

Notations

- For a sequence of sets, $A_1, A_2, \dots, A_n, \dots$,

$$\bigcup_{i=1}^n A_i \equiv \{x : x \in A_i \text{ for some } i = 1, 2, \dots, n\};$$

$$\bigcap_{i=1}^n A_i \equiv \{x : x \in A_i \text{ for each } i = 1, 2, \dots, n\};$$

$$\bigcup_{i=1}^{\infty} A_i \equiv \{x : x \in A_i \text{ for some } i \in \mathbb{N}\};$$

$$\bigcap_{i=1}^{\infty} A_i \equiv \{x : x \in A_i \text{ for each } i \in \mathbb{N}\}.$$

- Let X be a set and $\{A_x\}_{x \in X}$ a family of sets indexed by X .

$$\bigcup_{x \in X} A_x \equiv \{z : z \in A_x \text{ for some } x \in X\}.$$

$$\bigcap_{x \in X} A_x \equiv \{z : z \in A_x \text{ for each } x \in X\}.$$

Definition 2. Let X be a set. The power set of X , denoted $\mathcal{P}(X)$, is the set of subsets of X ,

$$\mathcal{P}(X) \equiv \{A : A \subseteq X\}$$

Alternatively $\mathcal{P}(X)$ is denoted 2^X .

The set of order pairs with the first component in A and the second component in B , denoted $A \times B$, is the set of lists of two elements, the first in A and the second in B , taking into account the order of the list. For instance if $A = \{1, 2\}$, and $B = \{1, 2, 5\}$, then $(1, 2)$, $(2, 5)$ are ordered pairs in $A \times B$. Note that $(2, 1)$ is an ordered pair in $A \times B$ too, and it is different than $(1, 2)$. Two ordered pairs (x_1, x_2) and (y_1, y_2) are equal if and only if $x_1 = y_1$ and $x_2 = y_2$. The set $A \times B$ is called the cartesian product between A and B . We summarize this in the following definition.

Definition 3. Let A, B be sets. The **cartesian product of A and B** , is the set

$$A \times B \equiv \{x : x = (a, b), a \in A, b \in B\}.$$

Example 6. $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$.

1.2.2 Functions

Set theory provides the right environment to study **functions**. Let A, B be two sets. Informally, a function f from A to B , is a *relation* such that for each $x \in A$ it assigns a unique point $f(x) \in B$. Observe that this definition is just informal given that we do not have a definition for *relation*. Nevertheless, we get the intuition that a function from a set A to a set B is a mathematical object that takes each element in A and attaches to it a unique element in B . If f is a function from A to B , we write $f : A \rightarrow B$. For each $x \in A$, the value of f at x is $f(x)$. In this case we say that f maps x into $f(x)$ and we write $x \xrightarrow{f} f(x)$.

Definition 4. (Technical) A **function** f from A to B is a subset of $A \times B$, such that for each $x \in A$, there is a unique $y \in B$, denoted $f(x)$, such that $(x, y) \in f$.

Definition 5. A **correspondence** from a set A to a set B is a function from A to 2^B . If Γ is a correspondence from A to B we write $\Gamma : A \rightrightarrows B$.

In the following definitions, $f : A \rightarrow B$.

Definition 6.

$D(f) \equiv A$ is the domain of f ; and

$R(f) \equiv \{y \in B : \exists x \in A, y = f(x)\}$ is the range of f .

Definition 7. Given a set $C \subset A$, the image of C under f is defined by

$$f(C) \equiv \{y \in B : \exists x \in C, y = f(x)\}.$$

Definition 8. Given a set $E \subset R(f)$, the inverse image of E under f is defined by

$$f^{-1}(E) \equiv \{x \in A : \exists y \in E, y = f(x)\}.$$

Definition 9. A function f is said to be **onto**, or **surjective**, if $\forall y \in B, \exists a \in A, f(a) = y$.

Definition 10. A function f is said to be **one-to-one**, or **injective**, if $\forall x, x' \in A, x \neq x' \Rightarrow f(x) \neq f(x')$.

A function that is **one-to-one and onto** is called **bijective**.

Definition 11. Given $f : X \rightarrow Y$ and $g : Y \rightarrow Z$, $g \circ f : X \rightarrow Z$ is defined by

$$g \circ f(x) = g(f(x)).$$